

# Model Performance Report

RecoMart Recommendation Pipeline | Task 9 & 10 Deliverable

## 1. Model Tracking Overview

All model training runs are tracked using MLflow. The MLflow tracking server records hyperparameters, evaluation metrics, and the trained model artifacts in a SQLite database (mlflow.db). This enables performance comparison across multiple training iterations.

## 2. Content-Based Recommendation Model

Algorithm: Gradient Boosting Regressor (Scikit-Learn)

Task: Predict the probability of a product being recommended based on product features.

### 2.1 Evaluation Metrics

| Metric   | Description                                              | Value   |
|----------|----------------------------------------------------------|---------|
| RMSE     | Root Mean Squared Error (Lower is better)                | 0.07536 |
| MAE      | Mean Absolute Error (Lower is better)                    | 0.04429 |
| R2 Score | Coefficient of Determination (Higher is better, Max 1.0) | 0.93862 |

Conclusion: The content-based model performs exceptionally well, explaining ~93.8% of the variance in recommendation probabilities.

## 3. Collaborative Filtering Model

Algorithm: Singular Value Decomposition (SVD) via Surprise Library

Task: Predict explicit user ratings for products using the user-item interaction matrix.

### 3.1 Evaluation Metrics

| Metric       | Description                                          | Value  |
|--------------|------------------------------------------------------|--------|
| RMSE         | Root Mean Squared Error (Rating scale 1-5)           | 1.0676 |
| MAE          | Mean Absolute Error                                  | 0.8705 |
| Precision@10 | Fraction of relevant items in top-10 recommendations | 0.000  |
| Recall@10    | Fraction of total relevant items retrieved           | 0.000  |
| NDCG@10      | Ranking quality score                                | 0.000  |

Conclusion: While the rating prediction error (RMSE ~1.06) is reasonable, the ranking metrics (Precision/Recall@10) are 0.0. This is expected due to the extreme sparsity of the synthetic interaction dataset (5,000 interactions spread across 200 users and 1,018 products). In a production setting with higher density data, these ranking metrics will become strictly meaningful.

## 4. Tracked Model Metadata

The following metadata is stored in MLflow for the latest run:

| Run Name                      | Run ID (Short) | Status   |
|-------------------------------|----------------|----------|
| collaborative_filtering_model | 568205e7       | FINISHED |
| content_based_model           | d03ee9ec       | FINISHED |